CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EC)

December 2012

EC 403 Mobile & Satellite Communication

Date: 05.12.2012, Wednesday **Time: 10.00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of a scientific calculator is allowed.

SECTION - I

Q - 1 (a) Draw a Timing Diagram illustrating how a call initiated by a mobile is established in Mobile Communication. **[04]**

(b) Explain Wireless Local Area Networks (WLANs) for Modern Wireless Communication Systems. **[03]**

Q - 2 (a) Prove that for a hexagonal geometry, the co-channel reuse ratio is given by $Q = \sqrt{3N}$, where $N = i^2 + ij + j^2$. **[06]**

(b) Illustrate a Handoff Scenario at cell boundary for Mobile Communication. **[04]**

(c) Explain Cell Splitting with necessary figure for Cellular System. **[04]**

OR

Q - 2 (a) Assume a receiver is located 10 km from a 50 W transmitter. The carrier frequency is 900 MHz, free space propagation is assumed, $G_t = 1$, and $G_r = 2$, find (i) The power at receiver (ii) The magnitude of the E-field at the receiver antenna (iii) The rms voltage applied to the receiver input assuming that the receiver antenna has a purely real impedance of 50 Ω and is matched to the receiver. **[06]**

(b) Describe signal prediction models like Okumura Model and Hata Model for urban area. **[04]**

(c) Explain Ground Reflection (Two-Ray) Model with necessary figure for Mobile Radio Propagation. **[04]**

Q - 3 (a) Compare Fixed Channel Allocation and Dynamic Channel Allocation in Cellular System. **[07]**

(b) Classify Small-Scale Fading based on multipath time delay spread and based on doppler spread. Elucidate fading effects due to Doppler Spread. **[07]**

OR

Q - 3 (a) What is Simple Borrowing Scheme in Cellular System? Explain Hybrid Channel **[07]**

Allocation, Flexible Channel Allocation and Handoff Channel Allocation.

(b) Define Co-Channel Interference. Compare the Co-channel interference with the Adjacent-channel interference [07]

SECTION – II

Q - 4 (a) Define Time Division Multiple Access (TDMA) System. Determine the Features of TDMA system. [04]

(b) Describe the functions of Base Station System and Mobile Switching Center for Digital Cellular Mobile System. [03]

Q - 5 (a) Draw the Block Diagrams of Quadrature Phase Shift Keying (QPSK) transmitter and receiver. Describe Offset QPSK with figures. [07]

(b) What is Inter Symbol Interference (ISI)? Draw the Block Diagram of a simplified communication system using an adaptive equalizer at receiver. Explain Basic Linear Equalizer during Training. [07]

OR

Q - 5 (a) Explain the "Call flow for authentication and ciphering" for Global System for Mobile Communication. Describe Modulation Process for IS-95 CDMA reverse traffic channels with necessary figures. [07]

(b) Define Vocoders. Compare the Frequency Diversity and the Time Diversity. [04]

(c) Explain Gaussian Minimum Shift Keying Modulation Technique for Mobile Radio. [03]

Q - 6 (a) Define Geostationary Satellite System. Explain Kepler's Laws with necessary figures for Satellite Communication. [07]

(b) Calculate the gain in decibels of a 4-m parabolic antenna operating at frequency of 12 GHz. Assume an aperture efficiency of 0.75. [04]

(c) Calculate the radius of a circular orbit for which the period is 1 day.The earth's geocentric gravitational constant equates $3.986005*10^{14}\, m^3/s^2$. [03]

OR

Q - 6 (a) Explain Different Types of Transmission Losses in the Space Link. [07]

(b) In Link Budget Calculation at 12 GHz for Satellite Communication System, the free-space loss is 210 dB, the antenna pointing loss is 2 dB, and the atmospheric absorption is 4 dB. The receiver [G/T] is 22.5 dB/K, and receiver feeder losses are 1 dB. The Effective Isotropic Radiated Power (EIRP) is 58 dBW. Calculate the carrier-to-noise ratio spectral density ratio. Boltzmann constant equates to -228.6 decilogs. [07]

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